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{
  "creator" : "ari.neko@cvut.cz",
  "timestamp" : "09-07-2026_17:24:43",
  "variant" : 3,
  "subjectName" : "mathematics",
  "categories" : [ "Probability", "Combinatorics", "Mathematical Logic" ],
  "questions" : [ {
    "question" : "What is the sum of probabilities of all possible mutually exclusive outcomes in a sample space?",
    "answers" : [ "Infinite", "0", "100", "1" ],
    "correctIndex" : 3
  }, {
    "question" : "What is the probability of rolling a sum of 7 with two fair six-sided dice?",
    "answers" : [ "1/6", "1/12", "1/36", "1/4" ],
    "correctIndex" : 0
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    "question" : "What is the probability of flipping a fair coin three times and getting all heads?",
    "answers" : [ "1/8", "1/6", "1/4", "1/2" ],
    "correctIndex" : 0
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    "question" : "How many unique permutations can be made from the letters in the word 'APP'?",
    "answers" : [ "6", "9", "1", "3" ],
    "correctIndex" : 3
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    "question" : "What is the value of the combinatorial coefficient  $C(5, 2)$ ?",
    "answers" : [ "20", "5", "15", "10" ],
    "correctIndex" : 3
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    "question" : "How many distinct subsets can be formed from a set containing 5 elements?",
    "answers" : [ "32", "25", "16", "64" ],
    "correctIndex" : 0
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    "question" : "In how many ways can 4 people be seated in a straight line?",
    "answers" : [ "16", "24", "6", "12" ],
    "correctIndex" : 1
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    "question" : "Which logical connective represents an 'exclusive or' (XOR)?",
    "answers" : [ "Exactly one input is true", "Both inputs are false", "Both inputs are true", "At least one input is true" ],
    "correctIndex" : 0
  }, {
    "question" : "What is the negation of the statement 'All primes are odd'?",

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"answers" : [ "No primes are odd", "All primes are even", "There exists at least one prime that is  
not odd", "Some primes are odd" ],  
"correctIndex" : 2  
}, {  
"question" : "If P is true and Q is false, what is the truth value of the implication  $P \rightarrow Q$ ?",  
"answers" : [ "False", "True", "Undefined", "Contingent" ],  
"correctIndex" : 0  
}]  
}
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